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Cell-specific roles of glucokinase in glucose homeostasis

C Postic [1], M Shiota, M A Magnuson

Affiliations
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Abstract

Mutations in the glucokinase (GK) gene cause two different diseases of blood glucose regulation: maturity onset diabetes of the young, type 2 (MODY-2) and persistent hyperinsulinemic hypoglycemia of infancy (PHHI). To gain further understanding of the pathophysiology of these disorders, we have used both transgenic and gene-targeting strategies to explore the relationship between GK gene expression in specific tissues and the blood glucose concentration. These studies, which have included the use of aCre/loxP gene-targeting strategy to perform both pancreatic beta-cell- and hepatocyte-specific knockouts of GK, clearly demonstrate multiple, cell-specific roles for this hexokinase that, together, contribute to the maintainance of euglycemia. In the pancreatic beta cell, GK functions as the glucose sensor, determining the threshold for insulin secretion. Mice lacking GK in the pancreatic beta cell die within 3 days of birth of profound hyperglycemia. In the liver, GK facilitates hepatic glucose uptake during hyperglycemia and is essential for the appropriate regulation of a network of glucose-responsive genes. While mice lacking hepatic GK are viable, and are only mildly hyperglycemic when fasted, they also have impaired insulin secretion in response to hyperglycemia. The mechanisms that enable hepatic GK to affect beta-cell function are not yet understood. Thus, the hyperglycemia that occurs in MODY-2 is due to impaired GK function in both the liver and pancreatic beta cell, although the defect in beta-cell function is clearly more dominant. Whether defects in GK gene expression also impair glucose sensing by neurons in the brain or enteroendocrine cells in gut, two other sites known to express GK, remains to be determined. Moreover, whether the pathophysiology of PHHI also involves multitissue dysfunction remains to be explored.

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